

1. A worker privately observes her type  $t \in [0, 1]$ , drawn randomly using the uniform distribution. After observing her type the worker may acquire education  $e \geq 0$  at a cost  $\frac{e}{t}$ .

Each firm values a type- $t$  worker with education  $e$  by  $f(t, e) = t^{\frac{1}{2}}$ . A worker with type  $t$  and education  $e$  that receives a wage  $w(e)$  has payoff

$$w(e) - \frac{e}{t}.$$

- (a) Please state whether the following statement is true or false and briefly explain your reasons.

An equilibrium requires that for every  $e$  the wage  $w(e)$  equals the firms' expected valuation of those workers who acquire education  $e$ .

- (b) Use the FOC of the worker's problem and part (a) to derive a differential equation that characterizes the wage function in a separating equilibrium, if it exists and  $w(e)$  is differentiable. (You need not solve the differential equation.)
- (c) Suppose  $w(e) = (3e)^{\frac{1}{3}}$ . What proportion of a type  $t$  value to the firm is spent on the costs of education?

2. Consider the following specification of the signaling model. A worker has ability  $\theta \in \{\theta_\ell, \theta_h\}$ ,  $0 < \theta_\ell < \theta_h$ , and chooses a level of education  $e \geq 0$  at a utility cost  $c(e, \theta) = k(\theta)z(e)$ , where  $k$  is strictly positive for all  $\theta$  and strictly decreasing in  $\theta$ , and  $z$  satisfies  $z(0) = z'(0) = 0$ ,  $z' > 0$  and  $z'' > 0$  for  $e > 0$ .

Ability is the worker's private information; the probability that the worker is of ability  $\theta_h$  (or the proportion of workers with  $\theta_h$  in the labor market) is  $q \in (0, 1)$ .

In turn, there are two firms that maximize expected profits and compete for the worker by posting wages after observing the education level chosen by the worker. A worker of ability  $\theta$  who chooses education  $e$  is worth  $\theta e$  to the firm (so if the firm pays  $w$  to this worker, then profit is  $\theta e - w$ ).

The timing of the game is: nature chooses  $\theta$ ; worker chooses  $e$  after learning his type; firms simultaneously post wages after observing  $e$ ; worker chooses which firm offer to accept, if any.

- (a) Define what a Weakly Perfect Bayesian Equilibrium (henceforth PBE) (in pure strategies) is in this game.
- (b) Construct a separating PBE involving the "least-costly" separation. Find the interval of education levels for  $\theta_h$  that can be part of a separating PBE.
- (c) Provide a verbalized but clear explanation of how notions of forward induction eliminate all PBE except for the "least-costly" separating PBE.  
Hint: Cho-Kreps (1987), Equilibrium Selection in Signaling Games, is helpful in answering this part of the question.

3. Consider the following competitive insurance market with adverse selection. There is a large number of identical insurance firms and heterogeneous consumers who have private information about their probability  $\theta \in [\underline{\theta}, \bar{\theta}] \subseteq [0, 1]$  of suffering a loss of size  $\ell \in (0, w)$ , where  $w$  is the initial

wealth of the consumers, and where  $\theta$  is distributed according to a cumulative distribution  $F$  that has a strictly positive density  $f$ . Insurance firms compete by offering full insurance (the policy coverage is the entire loss  $\ell$ ) at a price (premium)  $t$ . Consumers have a Bernoulli utility function  $u : \mathbb{R}_+ \rightarrow \mathbb{R}$  that is strictly increasing and strictly concave. If a consumer with type  $\theta$  does not buy insurance, then the consumer's expected utility is  $\theta u(w - \ell) + (1 - \theta)u(w)$ .

- (a) Define what a competitive equilibrium is in this market.
- (b) Prove that an equilibrium exists in this market.
- (c) Assume that  $f$  is uniform on  $[0, 1]$ . How many equilibria can there be? Show that there is always one in which the market breaks down. Assume in addition that  $u(c) = \log c$ ,  $w = 1$ ,  $\ell = 0.8$ , and show that there is an equilibrium with trade.
- (d) Suppose now there are only two types  $0 < \theta_l < \theta_h < 1$  and let  $\lambda \in (0, 1)$  be the fraction of high types in the market. Assume that there are two identical firms that compete by offering a finite number of contracts  $(x, t)$  where  $x$  is coverage, and consumers observe the contracts offered and choose the one that suits them best (or none of them). Set up the equations that describe the contracts that will be accepted by the two types of consumers in any subgame perfect equilibrium (in pure strategies) in this market, and draw a picture depicting these contracts (with  $t$  on the vertical axis and  $x$  on the horizontal one). In less than ten lines, explain how you would complete the equilibrium analysis of the screening game.